

Suraj Laddagiri

sladdagiri@ucsd.edu ❖ suraj@sladdagiri.org ❖ (424) 392-9264 ❖ Chino, CA ❖ San Diego, CA

Projects

Haptics And Vision for Environment Navigation (HAVEN)

- Led a multidisciplinary team of 4 to design and build a wearable navigation system for the visually impaired using iOS-based spatial computing
- Leveraged ARKit and RealityKit to implement real-time SLAM, utilizing iPhone LiDAR and cameras for semantic segmentation (detecting floors, walls, and doors)
- Developed a routing algorithm that translates 3D environmental data into directional haptic cues delivered via a custom tactor array
- Architected a custom hardware interface using an Arduino Uno R4 and an I2C multiplexer to independently control multiple haptic drivers

Music Strip

- iOS Application(Swift): Engineered a reactive dashboard to control LED hardware via CoreBluetooth, featuring dynamic state-driven navigation and real-time Spotify metadata integration
- Firmware Engineering (C++/ESP32): Developed multi-threaded firmware using FreeRTOS to handle high-priority lighting tasks, serial debugging, and BLE advertising simultaneously without latency
- Architected a custom communication protocol to synchronize hardware modes and animation parameters using string-based command parsing

iOS Systems Research & Reverse Engineering

- Conducted dynamic analysis of a large social media application using FLEX to map internal class hierarchies and reverse-engineer proprietary content delivery protocols
- Engineered a patch in Objective-C using the Logos hooking engine and Cydia Substrate through method swizzling to intercept and filter data streams
- Neutralized advanced multi-layer anti-tamper and jailbreak detection mechanisms through static analysis using Ghidra and patching ARM assembly instructions based on decompiled logic
- Decrypted and patched the binary to facilitate code injection on jailed iOS environments

EMPLOYMENT

University of California, San Diego

Jan 2024-Mar 2024

BENG 1 Instructional assistant

La Jolla, CA

- Designed several lab activities for the introductory Bioengineering class BENG 1 in conjunction with Dr. Robert Sah relating to Biomechanics, Bioelectricity, Bioinformatics, 3D Design and Modeling, and more.
- Guided students through labs, helping them understand material relevant to the topic being discussed
- Graded student submissions for accuracy and understanding of presented material

EXTRACURRICULAR ACTIVITIES

Improve the Dream

Oct 2022-Present

California Advocacy Team

Washington, DC

- Successfully urged the Biden administration to adjust age calculation under the CSPA, and introduce rulemaking for employment authorization for child dependents of visa holders at risk of aging out
- Corresponded with California governor Gavin Newsom's office and the Regents of the University of California regarding legislation and other changes conferring eligibility for AB540, In-State Tuition, and state student aid
- Met with both of California's senators, and UCSD's representative and lobbied them to co-sign the America's CHILDREN Act(S.2886/H.R.5528)

EDUCATION

University of California, San Diego

Sep 2022-June 2026

B.S, Bioengineering: Biotechnology & Mathematics-Computer Science, B.A, Political Science.

La Jolla, CA

- Relevant Courses: Computer Organization, Algorithms, Bioinformatic Algorithms, Data Structures, Linear Algebra, Numerical Analysis, Machine Learning, Natural Language Processing, Computer Vision

CERTIFICATIONS & SKILLS

- **Certifications:** Machine Learning(Andrew Ng DeepLearning.AI & Stanford University), Emergency Medical Technician(EMT-B) in The State of California(License # E184644)
- **Skills:** Swift, C, C++, Python, Assembly, Bioinformatics, Containerization, Machine Learning (PyTorch + TensorFlow)